AI-Powered Early Disease Risk Prediction and Remote Patient Monitoring System

Introduction

Many people get sick with problems like heart disease, diabetes, or high blood pressure without noticing the early warning signs. By the time they visit a doctor, the disease has often become serious and harder to treat. This project tries to solve this problem using simple wearable devices and computer programs (AI) that watch a person's health all the time and warn them early, before things get worse.

Problem Statement

Right now, most people only check their health when they visit a hospital or clinic once in a while. This is not enough, because health problems like heart disease and diabetes usually develop slowly over time. Small warning signs, such as a slightly high heart rate or unusual sugar levels, are easy to miss if nobody is watching regularly. This problem becomes even bigger for elderly people and people living in villages or remote areas, where hospitals are far away and regular check-ups are difficult. If there was a way to keep an eye on a person's health every day, doctors could catch problems early and treat them before they become dangerous. So, there is a real need for a system that can continuously check a patient's health, use AI to predict if something is going wrong, and send an alert to the patient, their family, or their doctor in time — all while keeping the patient's personal health data safe and private.

Abstract

This project is about building a smart health-monitoring system that uses wearable sensors and Artificial Intelligence (AI) to catch disease risks early and keep track of a patient's health from a distance. The system works like this: a wearable device (like a smartwatch or a small sensor kit) measures things like heart rate, blood pressure, oxygen level in blood (SpO2), and blood sugar level. This information is sent, through the internet or a mobile app, to a server where an AI model studies the data. The AI looks for patterns that might mean the person is at risk of a health problem, such as a heart issue or a diabetes complication, and gives the patient a "risk score." If the risk score becomes too high, the system automatically sends an alert to the patient, their caregiver, or a doctor, so that action can be taken quickly, before the situation becomes an emergency. The whole system has four main parts: 1. Sensors / Wearables — collect health data from the patient's body 2. Mobile App or Edge Device — sends the data to the cloud 3. AI Model in the Cloud — analyzes the data and predicts risk 4. Dashboard — shows the results and sends alerts to doctors or family members The goal of this project is to change healthcare from "treat the disease after it happens" to "prevent the

disease before it becomes serious," especially helping elderly patients and people who live far away from hospitals. The project also makes sure that patient data stays private and that patients agree (give consent) before their data is used.

Objectives

1. To collect real-time health data (heart rate, blood pressure, oxygen level, blood sugar) using wearable sensors. 2. To build an AI/Machine Learning model that can study this data and predict the risk of diseases like heart disease or diabetes at an early stage. 3. To design a mobile app or dashboard that shows the patient's health status in simple, easy-to-understand terms. 4. To create an alert system that automatically informs the patient, family members, or doctors when health risk becomes high. 5. To protect patient data through proper privacy and security measures, and take patient consent before using their data.

Expected Outcome

By the end of this project, we expect to have a working prototype that can track a person's basic health signals, predict early risk of disease using AI, and send timely alerts. This can help reduce emergency hospital visits, support elderly and rural patients, and show how AI and IoT together can make healthcare more proactive and accessible for everyone.

How the System Works (Simple Overview)

Think of it like a fitness band, but smarter and for health safety. Step 1 — Sensing: A small wearable device stays on the patient's body and keeps measuring things like heartbeat, blood pressure, oxygen level, and sugar level, just like a smartwatch tracks steps. Step 2 — Sending Data: This data travels from the wearable to a mobile phone app, and then to a secure server on the internet (called "the cloud"), where it can be safely stored and processed. Step 3 — AI Thinking: In the cloud, an AI model (trained earlier using lots of past patient health data) looks at the new data and compares it with known patterns of disease. It calculates a "risk score" — basically a number that tells how likely the person is to develop a health problem soon. Step 4 — Alerting: If the risk score crosses a certain limit, the system does not stay silent. It immediately sends a notification — a message or alert — to the patient's phone, and also to their family member or doctor, so someone can act quickly. Step 5 — Dashboard: Doctors or caregivers can log in to a simple dashboard (like an app or website) to see the patient's health history, current status, and past alerts, all in one place. In short: sensors collect data, the app sends it, AI studies it, and the system warns people early — all working together automatically, without the patient needing to do anything extra.

Relevance to Sustainable Development Goals (SDGs)

This project mainly supports Sustainable Development Goal 3: Good Health and Well-Being, because it helps people detect diseases early and get timely care, which can save lives and

reduce suffering. It also connects with Goal 10: Reduced Inequalities, since remote health monitoring can help people in villages and remote areas who do not have easy access to hospitals or doctors, giving them a similar level of healthcare monitoring as people in cities. Additionally, it touches Goal 9: Industry, Innovation and Infrastructure, as the project uses modern IoT (Internet of Things) and AI technology to build new, smart healthcare infrastructure.